

Answers:

1. (a) **Commutative laws:**

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

(b) **Associative laws:**

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

$$p \vee (q \vee r) \equiv (p \vee q) \vee r$$

(c) **Distributive laws:**

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

(d) **Identity laws:**

$$p \wedge t \equiv p$$

$$p \vee c \equiv p$$

(e) **Negation laws:**

$$p \vee \neg p \equiv t$$

$$p \wedge \neg p \equiv c$$

(f) **Double negative law:**

$$\neg(\neg p) \equiv p$$

(g) **Idempotent laws:**

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

(h) **Universal bound laws:**

$$p \vee t \equiv t$$

$$p \wedge c \equiv c$$

(i) **De Morgan's laws:**

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

(j) **Absorption laws:**

$$p \wedge (p \vee q) \equiv p$$

$$p \vee (p \wedge q) \equiv p$$

(k) **Negations of t and c :**

$$\neg t \equiv c$$

$$\neg c \equiv t$$

2. The Common Form should look like this.

Original Argument	Common Form
If I pass this final exam, then I won't be a disappointment to my family.	$p \implies \neg q$
I'm a disappointment to my family.	q
Therefore, it is not the case that I passed my final exam.	$\therefore \neg p$

3. Your truth table should look like this.

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$	$\neg(p \vee q) \wedge (\neg p \wedge \neg q)$
T	T	F	F	T	F	F	F
T	F	F	T	T	F	F	F
F	T	T	F	T	F	F	F
F	F	T	T	F	T	T	T

4. Is it logically equivalent?

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg p \vee \neg q$
T	T	F	F	T	F
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	F	T

False

5. Is it logically equivalent?

p	q	r	$q \wedge r$	$p \vee q$	$p \vee r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	T	F	F	F
F	F	T	F	F	T	F	F
F	F	F	F	F	F	F	F

True

6. Truth Table

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Tautology

7. $\neg(p \vee q) \wedge (\neg(\neg p \vee q)) \equiv c$

$\equiv (\neg p \wedge \neg q) \wedge (\neg(\neg p) \wedge \neg q)$	by De Morgan's laws
$\equiv (\neg p \wedge \neg q) \wedge (p \wedge \neg q)$	by Double negative law
$\equiv (\neg q \wedge \neg p) \wedge (\neg q \wedge p)$	by Commutative laws
$\equiv \neg q \wedge (\neg p \wedge p)$	by Distributive laws
$\equiv \neg q \wedge (p \wedge \neg p)$	by Commutative laws
$\equiv \neg q \wedge c$	by Negation laws
$\equiv c$	by Universal bound laws

8. Prove De Morgan's Laws by making a truth table.

(a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

(b) $\neg(p \vee q) \equiv \neg p \wedge \neg q$

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

9. Rewrite to this:

(a) $p \vee r$

(b) $p \wedge q$

(c) $(p \vee r) \wedge q$

(d) $(p \vee r) \wedge (q \vee s)$